

HealthTrack: Predicting Hospital Readmission

Abstract

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Hospital readmission within 30 days of discharge is one of the most significant quality indicators in modern healthcare. Unplanned readmissions not only impose an enormous financial burden on healthcare systems but also signal gaps in the quality of care provided during the initial hospital stay.

HealthTrack is a machine learning–based clinical decision support system designed to predict the likelihood of patient readmission within 30 days of discharge. The system is built using a Random Forest classifier trained on 2,000 synthetic patient records that capture key clinical and demographic features including age, gender, primary diagnosis, number of prior visits, length of hospital stay, type of discharge, insurance category, number of medications, and number of comorbid conditions.

The model achieves a test accuracy of 82% and an ROC-AUC score of 0.899, demonstrating strong discriminative ability between patients likely and unlikely to be readmitted. The system classifies patients into three risk tiers — Low, Medium, and High — allowing clinical teams to prioritize follow-up care for high-risk individuals before or immediately after discharge.

An interactive web application built using Streamlit provides an accessible interface for healthcare workers to enter patient information and receive an instant readmission risk estimate. The application also includes dataset insight charts and a full model evaluation report. This project demonstrates the practical application of supervised machine learning, data preprocessing, and web-based deployment in a healthcare context.

Keywords

Hospital Readmission, Random Forest, Binary Classification, Healthcare Analytics, Predictive Modeling, Streamlit, Python, Machine Learning